

INTRODUCTION TO EMBEDDED SYSTEM

An Embedded System is a specialized computer system designed to perform dedicated functions or tasks within a larger mechanical or electrical system. Unlike general-purpose computers (PCs), which are designed for a wide array of end-user needs, an embedded system is optimized for specific constraints such as power consumption, size, and real-time performance.

➤ Key Characteristics

1. **Purpose-Built:** The hardware is stripped of unnecessary components to save costs and space.
2. **Resource Constrained:** Operates with limited memory (RAM/Flash) and processing power (measured in MHz rather than GHz).
3. **High Reliability:** Designed to run for years without crashing or requiring a reboot (e.g., a pacemaker or an engine control unit).
4. **Efficiency:** Optimized for minimal power draw, often running on small batteries or energy harvesting.

➤ Real-World Applications

1. **Smart Home Systems**
 - a. Used in smart lights, ACs, security cameras, and door locks.
 - b. Helps in home automation and remote control.
2. **Automotive Systems**
 - a. Used in airbags, ABS, parking sensors, and engine control units.
 - b. Improves vehicle safety and performance.
3. **Healthcare Devices**
 - a. Used in ECG machines, heart rate monitors, insulin pumps, and pacemakers.
 - b. Helps in patient monitoring and treatment.
4. **Robotics**
 - a. Used in industrial robots, drones, and delivery robots.
 - b. Enables automation and accurate task performance.
5. **Industrial Automation**
 - a. Used in conveyor belts, machine control, and factory automation systems.
 - b. Increases productivity and reduces human effort.
6. **Consumer Electronics**
 - a. Used in washing machines, microwaves, smart TVs, and mobile phones.
 - b. Makes devices more efficient and user-friendly.
7. **Communication Systems**
 - a. Used in mobile phones, routers, and Wi-Fi devices.
 - b. Supports fast and reliable communication.
8. **Agriculture**
 - a. Used in smart irrigation systems and soil monitoring devices.

- b. Helps save water and improve crop production.

➤ **Difference Between Microcontroller And Microprocessor**

Microcontroller

- A microcontroller is designed for specific control-oriented tasks.
- It contains CPU, RAM, ROM, and I/O ports on a single chip.
- Used in embedded systems like washing machines and traffic lights.
- Consumes less power.
- Cost is lower.
- Suitable for small and dedicated applications.
- Processing speed is comparatively lower.
- Example: 8051, PIC, Arduino.

Microprocessor

- A microprocessor is designed for general-purpose computing tasks.
- It mainly contains only the CPU; memory and peripherals are external.
- Used in computers, laptops, and high-performance systems.
- Consumes more power.
- Cost is higher.
- Suitable for complex applications.
- Processing speed is higher.
- Example: Intel Pentium, AMD Ryzen.

COMPONENT STUDY

➤ **MICROCONTROLLERS**

Microcontrollers are small programmable devices that act like the “brain” of many electronic systems. They take input from sensors or buttons, process the information, and then control outputs such as LEDs, motors, displays, or alarms. They are widely used in embedded systems and automation projects.

1.Arduino

Arduino Uno is mainly used for learning and beginner-level electronics projects. It is easy to program and connect with sensors and modules. Arduino helps in controlling devices like LEDs, motors, and displays. It is commonly used in projects such as smart home systems, automatic street lights, and simple robots.

2.ESP32

ESP32 is a powerful microcontroller mainly used in IoT applications because it has built-in Wi-Fi and Bluetooth. It can connect devices to the internet and allow remote monitoring and control. ESP32 is used in smart home automation, wireless sensor systems, health monitoring devices, and smart agriculture projects.

3.Raspberry Pi Pico

Raspberry Pi Pico is a compact and low-cost microcontroller used for embedded and automation projects. It supports programming in MicroPython and C/C++. It is suitable for robotics, sensor interfacing, and industrial automation projects because of its fast processing and flexible input/output pins.

➤ BASIC SENSORS

Basic sensors are electronic components that detect changes in the environment and send signals to a microcontroller or system.

1. Temperature Sensor

A temperature sensor is used to measure temperature and detect changes in heat. It converts temperature into an electrical signal that can be processed by a microcontroller. These sensors are commonly used in weather monitoring systems, air conditioners, refrigerators, and healthcare devices.

Example: LM35- Used in temperature monitoring systems, air conditioners, and weather stations.

2. Motion Sensor

A motion sensor is used to detect movement of people or objects in a particular area. It is commonly used in security systems, automatic doors, street lights, and smart home

automation. When motion is detected, the sensor sends a signal to activate devices like alarms or lights.

Example: PIR Sensor - Used in automatic lights, security alarms, and smart home systems.

3. Light Sensor

A light sensor detects the intensity of light in the surroundings. It helps systems automatically respond to changes in brightness. Light sensors are used in automatic street lights, mobile screen brightness adjustment, and smart lighting systems.

Example: LDR - Used in automatic street lights and brightness control systems.